

Multichannel Anomaly Detection for Spacecraft Time Series Using MAP Estimation

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Automated anomaly detection in spacecraft telemetry systems is essential for analyzing abnormal events and system failures. A widely adopted strategy is to predict the target time sequences using a machine learning method first, then extract the anomalies from the residuals between the target time sequences and the predicted sequences by a thresholding method. Although thresholding-based anomaly extraction is simple and fast, it fails to take advantage of correlations between anomaly sequences over time and across channels. To make the process of anomaly extraction more flexible and more accurate, a statistical model referred as an anomaly marked point process (Anomaly-MPP) is proposed in this article. This model treats anomaly sequences as objects to be detected, making the anomaly detection a classical object detection problem. Formulating this as an optimization problem, we find the maximum a posteriori estimate of the set of anomaly objects in a multichannel time-series dataset, modeling the prediction error sequences generated from the output of a transformer with the proposed Anomaly-MPP for the posterior distribution. The prior distribution can incorporate domain knowledge and user-specified context into the problem formulation, thus providing additional detection “power.” By including a length prior energy term and a correlation prior energy term into the model, the

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anomaly extraction process not only considers the prediction error values, but also takes the length of detected anomaly sequences and the interchannel dependencies into account. A case study is given in the experimental section to illustrate the use of the model on a real dataset. Also, the effectiveness of our method is evaluated on an Mars Reconnaissance Orbiter dataset with inserted known anomalies and two public datasets: Secure Water Treatment and Water Distribution.

I. INTRODUCTION

There are thousands of sensors in a typical spacecraft for collecting various information about the state of the spacecraft [1], such as temperature, power consumption, and wheel speed. This information can be used to determine the status of the spacecraft and help experts to understand the reasons for a system failure. Usually, subject matter experts utilize their domain-based knowledge to analyze subsystem components during root cause analysis by inspecting abnormal events from downlink telemetry [2]. However, massive amounts of time-series data are collected every day, and it is extremely time-consuming to analyze all the data manually. Hence, a dependable automated anomaly detection algorithm is essential for reducing the burden on domain experts.

Due to the lack of labeled anomalous data, anomaly detection in time series is often realized in a semisupervised way. Such semisupervised anomaly detection approaches can learn the normal and expected behavior of a telemetry channel, so the anomalies can be identified by the deviations from this behavior [3]. To learn the normal behavior of a time series, there are mainly two kinds of models: prediction-based learning models and reconstruction-based learning models.

A prediction-based model usually learns to predict the unknown data in the future from past known data. Traditionally, an autoregressive integrated moving average model [4] has been a common choice for time series prediction. However, it makes a stationarity assumption on the data that may not be satisfied in a spacecraft system. Support vector machines are also a potential choice for prediction [5], [6], and their use in anomaly detection in time series has been demonstrated in [7]. The ability to learn long patterns and contextual reasoning makes long short-term memorys (LSTMs) [8], [9] a popular model for anomaly detection in time series [10], [11], [12]. They have also proven to be effective at adapting to new data via transfer learning [13]. Ahmad et al. [14] proposed the hierarchical temporary memory for online anomaly detection without labeled data. Baireddy et al. [15] applied extreme learning machines to real-time anomaly detection. With the successful application of transformers [16], [17] in natural language processing and time series forecasting [18], [19], their potential for anomaly detection has been demonstrated in [20] and [21].

To fully use the spatial and temporal information of multichannel time series for predicting the unknown future behavior, Tian et al. [22] proposed an anomaly detection network using spatial and temporal information, which models the relationship graph between variables for a graph

attention network to capture the spatial dependency between variables and utilizes an LSTM to mine the temporal dependency of time series. Due to the usefulness of graphs in depicting objects and their relationships, several studies have utilized dynamic graphs [23], [24], [25] to represent multivariate time series and their correlations over continuous time intervals. Examples include the multivariate time-series anomaly detection via graph attention network proposed by Zhao et al. [23] and the multimodal spatial-temporal graph attention network proposed by Ding et al. [25].

A reconstruction-based model represents the input data by some latent variables first, then reconstructs the input data from such latent variables. In [26], generative adversarial networks [27] were proposed as a reconstruction model for anomaly detection. In [28], the OmniAnomaly detector is proposed to use a stochastic recurrent neural network to reconstruct the input data and determine anomalies according to the reconstruction probabilities. Given the capability of the variational autoencoder model (VAE) [29] to learn a low-dimensional latent space representation of input time series, Lin et al. [30] combined VAE with an LSTM model for input data reconstruction in anomaly detection. To further learn the long-term dependence of time series data, Chen et al. [31] proposed an RT-Semi VAE, which combines transformer and VAE.

No matter whether the learning model is prediction based or reconstruction based, for each target sequence, it is necessary to generate a residual sequence by deducting the prediction/reconstruction sequence from the target sequence. Then, an anomaly extraction algorithm will be applied to the residual sequences to identify the anomaly sequences. In [10], the anomalies are extracted from the prediction error sequences by a dynamic thresholding algorithm. In our previous work [2], instead of extracting the anomalies from the prediction error sequence directly, we model the distribution of error values by kernel density estimation [32] and convert the error sequence to a likelihood value sequence based on the error distribution first, then extract the anomalies from the likelihood value sequence by a dynamic thresholding method. A peaks-over-threshold (POT) model [33] is another choice for anomaly extraction from prediction error sequences, having been applied in [21], [23], [28], and [34].

Thresholding-based anomaly extraction algorithms work fine for single-channel time series when the learning model is reliable (the prediction sequence matches the normal part of the target sequence well and the abnormal part of the target sequence poorly). However, when the performance of the learning model is not sufficient or there is a lot of noise in the target sequence, the extraction results can be poor. Moreover, for the case of multichannel time series, the single-channel thresholding method fails to consider channel dependencies, rendering it inadequate for anomaly detection in spacecraft time series as anomalies in different channels could be related.

To make the anomaly extraction process more flexible and robust for spacecraft time series, we propose an

anomaly marked point process (Anomaly-MPP) to model the anomaly sequence objects in the target sequences. The MPP objects in this case are the anomaly sequences. To fit an anomaly object with the prediction error sequences, a data energy term is defined based on the contrast between the prediction errors in a region and those in its neighboring regions. Also, a length prior term is proposed to penalize short anomaly sequence objects, which are likely to be noise. Moreover, a correlation prior term is proposed in our model, allowing us to identify an anomaly object with the information of detected anomaly objects in all other channels. Note that very short anomalies and single-channel anomalies might still be detected as detection depends on the complete posterior distribution.

Considering the ability of a transformer to draw global dependencies between input and output through its attention mechanism [16], we use a bidirectional transformer-based predictor to learn the normal behavior from time series and generate the predicted sequences. For the task of anomaly detection in multiple-channel time series, a residual sequence is calculated by subtracting the prediction sequence from the target sequence for each channel. Then, the proposed Anomaly-MPP model is used for the residual sequences, and maximum a posteriori (MAP) estimation is used for anomaly extraction. Note that the novelty in this article is in the modeling and optimization of the multichannel prediction error sequences, and although we use a transformer in this work, the method proposed here could be used with other prediction/reconstruction approaches.

In summary, our work makes the following contributions.

- 1) We propose a novel statistical model Anomaly-MPP with MAP estimation for anomaly extraction in multichannel time series.
- 2) We introduce the length prior energy term and the correlation prior energy term into our model to suppress short anomaly objects and establish relationships among anomaly objects across different channels, which is essential for anomaly detection in spacecraft time series.
- 3) Extensive experiments show the flexibility and effectiveness of our approach, outperforming the commonly used thresholding-based methods.

The rest of this article is organized as follows. In Section II, the problem of anomaly detection is properly defined. In Section III, we briefly review the structure of the transformer and how we use it in our application first, then the Anomaly-MPP model is presented in detail. A standard method for MAP optimization for MPPs is used, so we describe it briefly. In Section IV, experimental results are given to show the performance of our approach. Finally, Section V concludes this article.

II. PROBLEM STATEMENT

In this work, we address the problem of anomaly detection in multichannel time series. Denote the target time

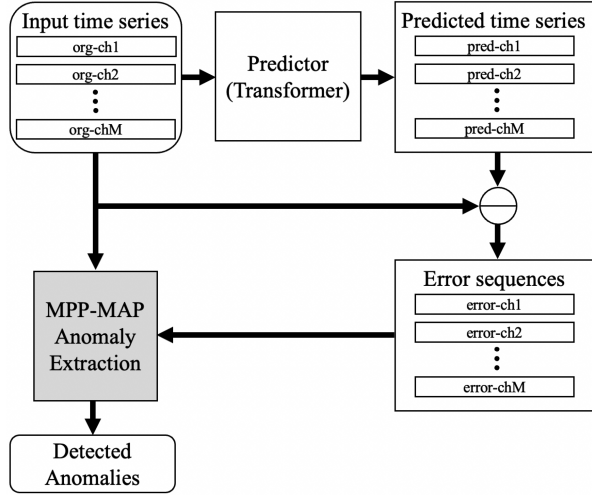


Fig. 1. Diagram of the proposed anomaly detection method. The gray block is the novel contribution of this article.

series as $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N]$, where $\mathbf{x}_t \in \mathbf{R}^M$, N is the number of samples in $\mathbf{X}$, and M is the total number of channels.

In a spacecraft system, a system-level anomaly may trigger anomalies in several channels, and these anomalies may not always occur at the same time. For example, an anomaly could occur in an electrical power channel before a corresponding anomaly appears in a related temperature channel. To assist with root cause analysis, our objective is not only to answer the question of when an anomaly occurs but also to give information about which channels are questionable. That is, for the target time series $\mathbf{X} \in \mathbf{R}^{M \times N}$, our objective is to find a set of anomaly sequences $[(i, k_i, \text{start}_i, \text{end}_i)]$, where i means the i th anomaly, k_i is the channel where the anomaly is found, and start_i and end_i are the starting and ending time of the i th anomaly, respectively.

III. METHODOLOGY

Fig. 1 shows the diagram of our method for multichannel anomaly detection. There are mainly two parts: the first is the time series predictor, which is a pure transformer-based predictor. We train the predictor with sequences assumed to have no anomalies. Since the spacecraft system runs at a normal status most of the time, such normal data are easily accessible. In the anomaly detection process, we feed the target multichannel sequences into the trained predictor to generate the predicted sequences. An error sequence is computed as the difference between a target sequence and its predicted sequence. The second part of our approach is an anomaly extractor that uses MAP estimation to detect anomaly objects from the error sequences, which are modeled using an Anomaly-MPP. With the data energy, length prior energy, and correlation prior energy defined in the Anomaly-MPP model, it is flexible to extract the anomaly objects with different settings. The data energy promotes the detection of objects with high error contrast. Meanwhile, the length prior energy discourages the identification of short objects with low error contrast while favoring those with

greater lengths. In addition, the correlation prior energy incorporates insights from other channels.

In this section, we describe the structure of the transformer and introduce how we use it in our work first, then the proposed Anomaly-MPP model is discussed in detail. Finally, we give a brief description of the MAP optimization method we use.

A. Transformer Review

Our choice of a transformer-based predictor stems from its exceptional capabilities in capturing temporal dependencies and modeling complex sequences. Unlike traditional sequence models, such as recurrent neural networks, transformers leverage self-attention mechanisms to efficiently capture long-range dependencies across sequences. This ability is crucial for our task, which involves predicting data points where signals may exhibit complex temporal patterns. In addition, transformers offer scalability and parallelization, enabling efficient processing of large-scale sequential data without sacrificing model performance.

The transformer-based predictor utilized in our method follows the original transformer architecture [16]. The main differences between the transformer-based forecasting model used in [18] and our predictor are the input and output. Since our task is neither real-time anomaly detection nor forecasting of unknown future data, we do not use the left-to-right forecasting model. Instead, a bidirectional prediction model is used, as shown in Fig. 2. Intuitively, a bidirectional model is more powerful [17], especially for predicting the data at points where a signal changes abruptly.

For a subsequence $[\mathbf{x}_{t-L}, \mathbf{x}_{t-L+1}, \dots, \mathbf{x}_t, \dots, \mathbf{x}_{t+L}]$ of the test sequence $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N]$, where L controls the size of the transformer input. We mask the middle three data samples with 0 s to generate the input data for time t : $\mathbf{X}_t = [\mathbf{x}_{t-L}, \dots, \mathbf{x}_{t-2}, \mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{x}_{t+2}, \dots, \mathbf{x}_{t+L}]$. We try to predict $\mathbf{x}_t$ based on the input $\mathbf{X}_t$. The reason for masking the middle three data samples rather than only masking $\mathbf{x}_t$ is to avoid the trivial prediction of replicating the immediately preceding or succeeding values or by the average of the two [35].

The transformer has an encoder-decoder architecture. The encoder encodes the input representation $\mathbf{X}_t$ into a hidden state representation $\mathbf{Z}_t = [\mathbf{z}_{t-L}, \mathbf{z}_{t-L+1}, \dots, \mathbf{z}_t, \dots, \mathbf{z}_{t+L}]$, where $\mathbf{z}_i \in \mathbf{R}^{d_{\text{model}}}$, and d_{model} is the model dimension. The decoder decodes an output representation $\mathbf{A}_t = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_{L_d}]$ from the hidden representation $\mathbf{Z}_t$ and the rightmost L_d elements of $\mathbf{X}_t$, where L_d is the size of the input of the decoder, $\mathbf{a}_i \in \mathbf{R}^{d_{\text{model}}}$. Finally, a linear layer is applied to map $\mathbf{A}_t$ to the prediction $\hat{\mathbf{x}}_t$.

The encoder is composed of an input layer, a positional encoding layer, and a stack of four identical encoder layers. The input layer linearly projects the M -dimensional input onto a d_{model} -dimensional vector space. Since the attention model contains no recurrence and no convolution operations, to make use of the order information of the sequence, the positional encoding layer is applied to inject position information with sine and cosine functions [16]. The encoder

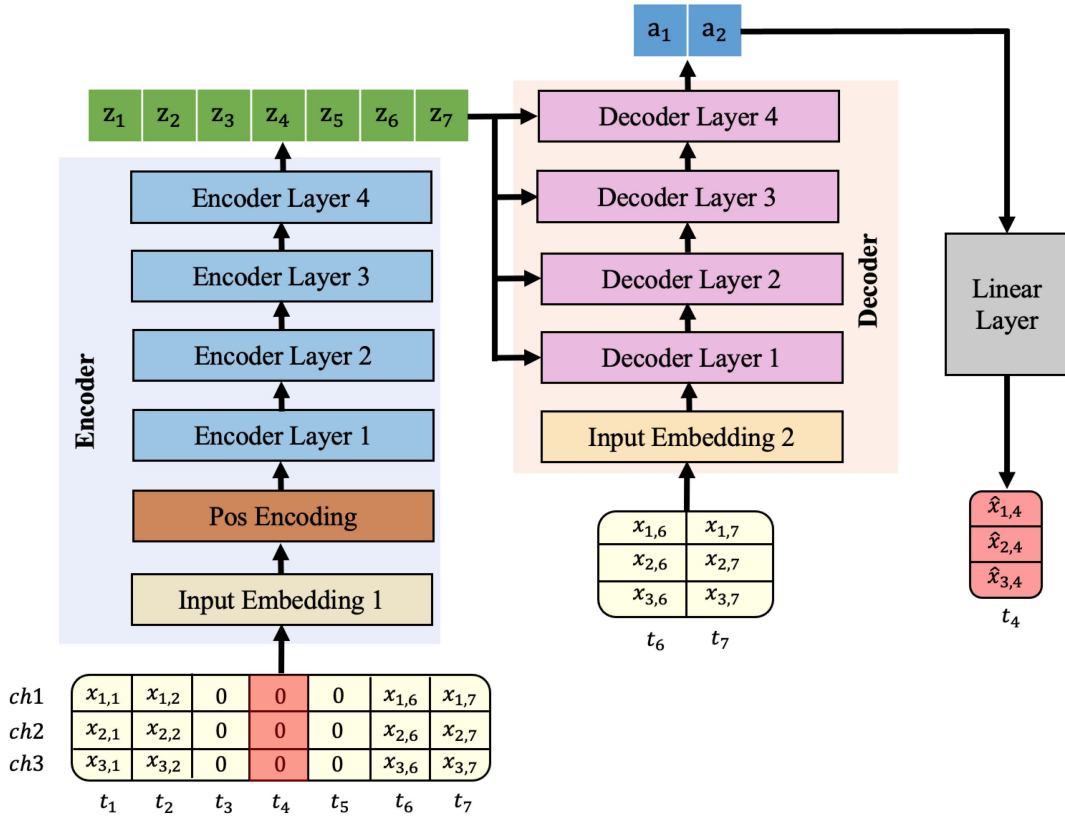


Fig. 2. Structure of the bidirectional transformer-based predictor with $L = 3$ and $L_d = 2$.

layer uses one multi-attention block to learn the dependencies of data with different positions. A multi-attention block consists of six attention blocks, each attention block realizing the “scaled dot-product attention” by

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

where Q, K , and V are the query matrix, key matrix, and value matrix, respectively, $Q \in R^{q \times d_k}$, $K \in R^{p \times d_k}$, $V \in R^{p \times d_v}$, q is the number of queries, p is the number of key-value pairs, d_k is the dimension of each key and query, and d_v is the dimension of each value.

The decoder is composed of an input layer and a stack of four identical decoder layers. The input layer is the same as the one in the encoder. The decoder layer has two multi-attention blocks, one for learning the dependencies of input data and the other for integrating the information from the hidden representation Z_I . To read the details of the encoder layer and the decoder layer, see the original transformer paper [16].

The commonly-used mean squared error loss function is used to train the transformer-based predictor in our application.

B. Anomaly-MPP Model

With the transformer trained on the training data, we can generate a predicted sequence $\hat{X} = [\hat{x}_{L+1}, \hat{x}_{L+2}, \dots, \hat{x}_{N-L}]$

from the test sequence $X = [x_1, x_2, \dots, x_N]$, where $\hat{x}_n = [\hat{x}_{1,n}, \hat{x}_{2,n}, \dots, \hat{x}_{M,n}]$ and $x_n = [x_{1,n}, x_{2,n}, \dots, x_{M,n}]$.

Let $E = [e_1, e_2, \dots, e_N]$ be the absolute error sequence, where $e_n = [e_{1,n}, e_{2,n}, \dots, e_{M,n}]$ and $e_{m,n} = |x_{m,n} - \hat{x}_{m,n}|$ (Note: for convenience, we ignore the first and last L data points in the test sequence). When $e_{m,n}$ is relatively large, it indicates potential abnormal behavior at time n in channel m .

The Anomaly-MPP is proposed to model the set of anomalies in the multichannel input based on the error sequence E . In this section, we will define the Anomaly-MPP model first, then discuss the data and prior energy terms of the model, and finally, briefly describe the optimization method we use to find the MAP estimate of the set of anomaly sequences.

1) *Anomaly-MPP Model:* Let S be the $M \times N$ lattice, $S = [1, M] \times [1, N] \subset R^2$, upon which the input data X are defined. Then, a point process on S is a set of points $\{S_1, S_2, \dots, S_K\} \subseteq S$, with random variables $S_i = (m_i, n_i)$ representing the random location of the i th point, where m_i is its channel identifier (ID) and n_i is the time location. The random variable K is the total number of points in the point process [36], [37].

By associating a mark b_i to each point object S_i , we get an MPP [38], [39]. In the Anomaly-MPP model, $b_i \in B$ is the random width of the i th anomaly object, where $B = [b_{\min}, b_{\max}]$ is the mark space with parameters $b_{\min}$ and $b_{\max}$. A marked object is denoted as $W_i = (S_i, b_i) \in S \times B$. Then, an anomaly configuration is defined as $W =$

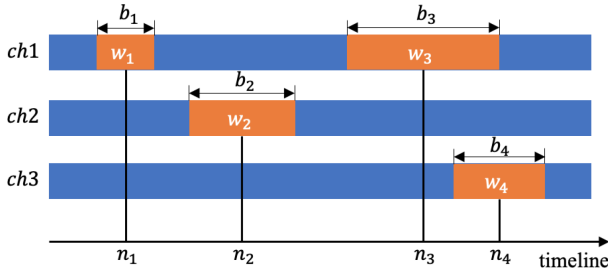


Fig. 3. Example of anomaly object configuration.

$\{W_1, W_2, \dots, W_K\}$. The space of all possible configurations Ω_W has elements of the form $\mathbf{w} = \{w_1, w_2, \dots, w_k\}$, containing anomaly objects $w_1, \dots, w_k$. Fig. 3 shows an example of an object configuration within three channels.

A Gibbs model is used for the MPP, which results in a posterior density function given by

$$f(\mathbf{w}|\mathbf{e}) = \frac{1}{Z} \exp\{-V_d(\mathbf{e}|\mathbf{w}) - V_p(\mathbf{w})\} \quad (2)$$

where $\mathbf{w}$ is an anomaly object configuration, $\mathbf{e}$ is the observed error sequence, $V_d(\mathbf{e}|\mathbf{w})$ is the data energy, which determines the fit of the anomaly configuration $\mathbf{w}$ to the observed error sequence $\mathbf{e}$, $V_p(\mathbf{w})$ is the prior energy, and Z is the normalizing constant. Our goal is to find the object configuration that maximizes $f(\mathbf{w}|\mathbf{e})$, that is, the MAP estimate of $\mathbf{W}$ given that the observation error sequence is $\mathbf{e}$.

It remains to define the energy functions V_d and V_p in (2). We will use a common “reward” function [40] [given in (6)] for V_d and for each part of V_p , as described next.

2) *Data Energy*: Under the assumption that the elements of the error sequence E are conditionally independent given that $\mathbf{W} = \mathbf{w}$ for any configuration $\mathbf{w} \in \Omega_W$, the data energy term of (2) can be written as

$$V_d(\mathbf{e}|\mathbf{w}) = \sum_{i=1}^k \tilde{V}_d(\mathbf{e}|w_i) \quad (3)$$

where k is the number of anomalies in configuration $\mathbf{w}$. The basic idea of the data energy we propose is this: if the mean error value of object w_i is larger than the mean error value of its neighboring regions, then it has higher probability to be a true anomaly object, and thus should have a smaller $\tilde{V}_d(\mathbf{e}|w_i)$.

Define the outer area of object w_i as the union of its left neighborhood left_i and right neighborhood right_i , which are the green areas shown in Fig. 4. We set the neighborhood area length to $b_{\max}$, where $b_{\max}$ is the parameter of the maximum object width. Then, let $u_{\text{in}}(w_i)$ be the mean error value of object w_i , and $u_{\text{out}}(w_i)$ be the mean error value of its outer area. Define the error contrast as

$$\text{contrast}(w_i) = \frac{u_{\text{in}}(w_i) - u_{\text{out}}(w_i)}{u_{\text{out}}(w_i)}. \quad (4)$$

Let $D(w_i) = G_d \text{contrast}(w_i)^2$ with data gain G_d , being fixed for our experiments, and set

$$\tilde{V}_d(\mathbf{e}|w_i) = h(D(w_i), G_d T_d^2, K_d) \quad (5)$$

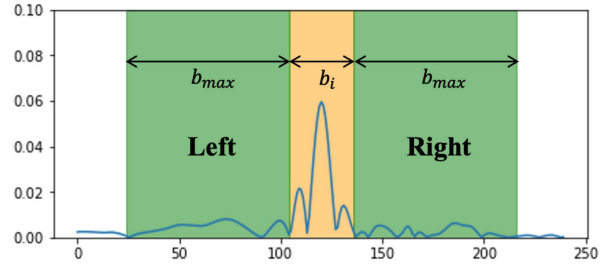


Fig. 4. Inner area (yellow) and outer area (green) of an object.

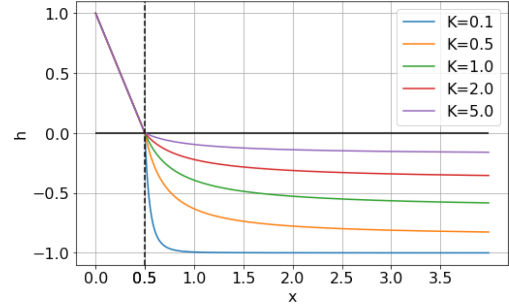


Fig. 5. Function of $h(x, T, K)$ with $T = 0.5$ and $K = 0.1, 0.5, 1, 2, 5$, respectively.

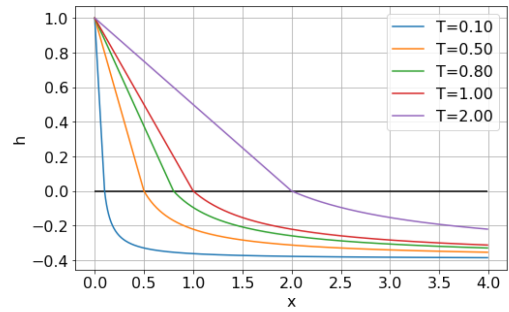


Fig. 6. Function of $h(x, T, K)$ with $K = 2$ and $T = 0.1, 0.5, 0.8, 1, 2$, respectively.

where the data term threshold T_d and data reward coefficient K_d are parameters settable by the user, and h is the reward function defined as

$$h(x, T, K) = \begin{cases} 1 - \frac{x}{T}, & \text{if } x < T \\ \exp\left\{\frac{-(x-T)}{Kx}\right\} - 1, & \text{else.} \end{cases} \quad (6)$$

Figs. 5 and 6 present the function h with fixed T and fixed K , respectively. As we can see, when $x < T$, the function returns a linear positive value, which can be taken as a linear penalty. When $x > T$, the function returns a negative reward, which is controlled by the reward coefficient K .

3) *Prior Energy*: The prior energy $V_p(\mathbf{w})$ describes prior information about objects, and is given by

$$V_p(\mathbf{w}) = V_p^{\text{ol}}(\mathbf{w}) + \alpha V_p^{\text{len}}(\mathbf{w}) + \beta V_p^{\text{co}}(\mathbf{w}). \quad (7)$$

The overlap prior $V_p^{\text{ol}}(\mathbf{w})$ rejects overlapping between any two objects in the same channel, the length prior $V_p^{\text{len}}(\mathbf{w})$ penalizes objects with short length, the correlation prior $V_p^{\text{co}}(\mathbf{w})$ takes account of the relationship among objects in



Fig. 7. Overlapping of two objects.

different channels, and α and β are the weights for the last two terms.

a) *Overlap prior*: For the task of anomaly detection, we do not allow overlapping between any two objects within a channel. That is if two objects are in the same channel and overlap with each other, as shown in Fig. 7, then the one with larger data energy should be removed. This helps to avoid the detection of a single anomaly as multiple different anomalies. Define

$$V_p^{\text{ol}}(\mathbf{w}) = \sum_{i \neq j} g(w_i, w_j) \quad (8)$$

where

$$g(w_i, w_j) = \begin{cases} \infty, & \text{if } w_i \text{ and } w_j \text{ overlap} \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

b) *Length prior*: To avoid false alarms caused by noise, the length prior term $V_p^{\text{len}}(\mathbf{w})$ is introduced to encourage long objects and penalize short objects. We let

$$V_p^{\text{len}}(\mathbf{w}) = \sum_{i=1}^k \tilde{V}_p^{\text{len}}(w_i) \quad (10)$$

where

$$\tilde{V}_p^{\text{len}}(w_i) = h(L(b_i), L(T_l), K_l). \quad (11)$$

In this case

$$L(x) = G_l R_l(x)^2 \quad (12)$$

and

$$R_l(x) = \frac{x - b_{\min}}{b_{\max} - b_{\min}} \quad (13)$$

so the first argument to the reward function h is a quadratic function of the length of anomaly i , the second argument is the same quadratic function of the length threshold, and the third argument is the length reward coefficient. The length gain G_l is fixed for our experiments.

c) *Correlation prior*: The correlation prior $V_p^{\text{co}}(\mathbf{w})$ is introduced to model the relationships among objects in different channels. Fig. 8 gives an example to illustrate the basic idea of the correlation prior. For any two channels, say channel m_1 and channel m_2 , suppose that an object w_1 is in channel m_1 at location n_1 , and another object w_2 exists in channel m_2 at location n_2 . We expand w_1 and w_2 by a width d , as shown in the green area in Fig. 8. If the expanded object w_1 and object w_2 are overlapped in the timeline, then both of them should have a higher probability to be true anomalies, thus we reduce the prior energy for w_1 and w_2 .

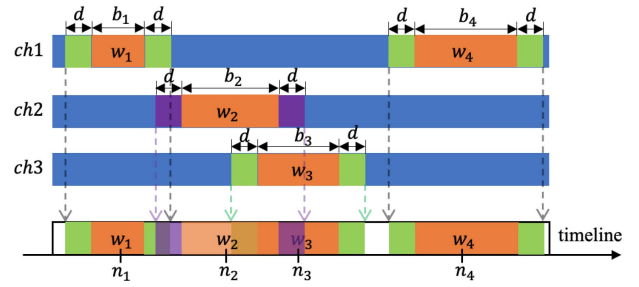


Fig. 8. Illustration for correlation prior.

To simplify computation, we compute the correlation prior for each object separately and let

$$V_p^{\text{co}}(\mathbf{w}) = \sum_{i=1}^k \tilde{V}_p^{\text{co}}(w_i). \quad (14)$$

For object w_i in channel m_i located at time n_i , we count the number c_i of channels that have expanded objects, which overlap with the expanded w_i in the timeline. Then, we let

$$\tilde{V}_p^{\text{co}}(w_i) = h(P(c_i, M), P(T_{\text{co}}, M), K_{\text{co}}) \quad (15)$$

where

$$P(x, M) = G_{\text{co}} R_2(x, M) \quad (16)$$

and

$$R_2(x, M) = \frac{x}{M - 1} \quad (17)$$

where T_{co} is the correlation threshold parameter, K_{co} is the correlation reward coefficient, and G_{co} is the correlation gain, which is fixed for our experiment.

4) *Optimization Method*: To find the MAP estimate of the true anomaly configuration under our model, the Gibbs energy in (2) must be minimized, so that the posterior distribution is maximized. A reversible jump Markov chain Monte Carlo [41] method is widely used for this type of problem when the normalizing constant of the posterior density is difficult to calculate. However, this algorithm is mainly based on proposing local perturbations, which limits its convergence speed. Instead, we use the multiple birth and death (MBD) algorithm [42], [43], which is able to make multiple perturbations in parallel, to search for the optimal configuration.

The MBD algorithm runs a birth step and a death step iteratively. At iteration i , given the previous configuration $\mathbf{w}(i-1)$, we propose a new configuration $\mathbf{w}' = \{w'_1, w'_2, \dots, w'_k\}$ to get $\mathbf{w} = \mathbf{w}(i) \cup \mathbf{w}'$ in the birth step. In the death step, we update $\mathbf{w}(i)$ by removing nonfitting objects from $\mathbf{w}$. The algorithm keeps iterating until a convergence condition is satisfied or the maximum iterations set by the user is reached.

IV. EXPERIMENTS

To illustrate the proposed Anomaly-MPP-MAP approach in detail, a case study is made on five selected channels from the Soil Moisture Active Passive satellite (SMAP) dataset [44]. Then, we test the method on a Mars

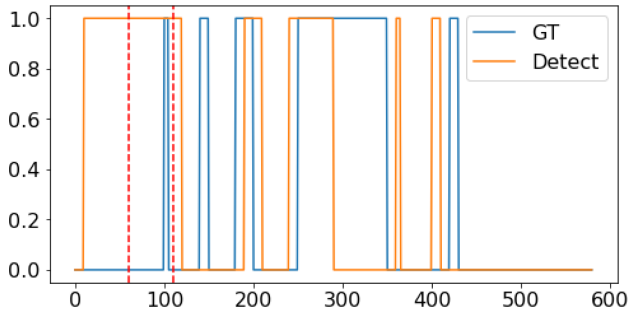


Fig. 9. Example to compare different anomaly counting methods.

Reconnaissance Orbiter (MRO) dataset, to which we have inserted multichannel anomalies, to show its advantage over threshold-based methods for anomaly extraction in multichannel series. Finally, we apply the Anomaly-MPP-MAP approach to a single accumulated error sequence generated from Secure Water Treatment (SWaT) and Water Distribution (WADI) datasets [45], [46] to further verify its effectiveness.

A. Datasets

Four datasets have been used in our experiments: SMAP dataset [10], [44] which is a spacecraft dataset collected by NASA; MRO dataset, which is a spacecraft dataset provided by Lockheed Martin; SWaT dataset [45], which was collected from a water treatment testbed for cyber-attack investigation in 2015; and WADI dataset [46], which was collected from a water distribution testbed as an extension of the SWaT testbed.

The SMAP dataset contains 54 channels of time series, each with 25 features (the first feature is the telemetry value and others are related to system status). There is a training sequence and a test sequence for each channel, with the test sequences varying in length. For our illustrative case study, we select the telemetry value sequences from five channels (A-2, A-3, A-4, E-1, and E-10) with similar length to form a multichannel test case, which is partially shown in Fig. 10. There are a total of seven anomaly sequences in these channels.

The MRO dataset includes thousands of channels, which encompass all spacecraft subsystems, and is downlinked to a ground system at varying sample rates [2]. The MRO dataset does not contain ground truth anomalies and is considered an unlabeled dataset. For each channel, a single week of data preceding a system reboot is given. We assume that the first five-and-half days of data of the time series in each channel represents normal functioning. By this assumption, eight groups of data were created from the first five days of data (the original data points were downsampled to one measurement every 30 s). In each group, ten channels were randomly selected from the MRO dataset. The time series is divided into training sequences and test sequences for each channel, with lengths of 5000 and 10 000, respectively. Anomalies were injected into the test sequence in a similar way as described in [47]. There are a total of 140 injected

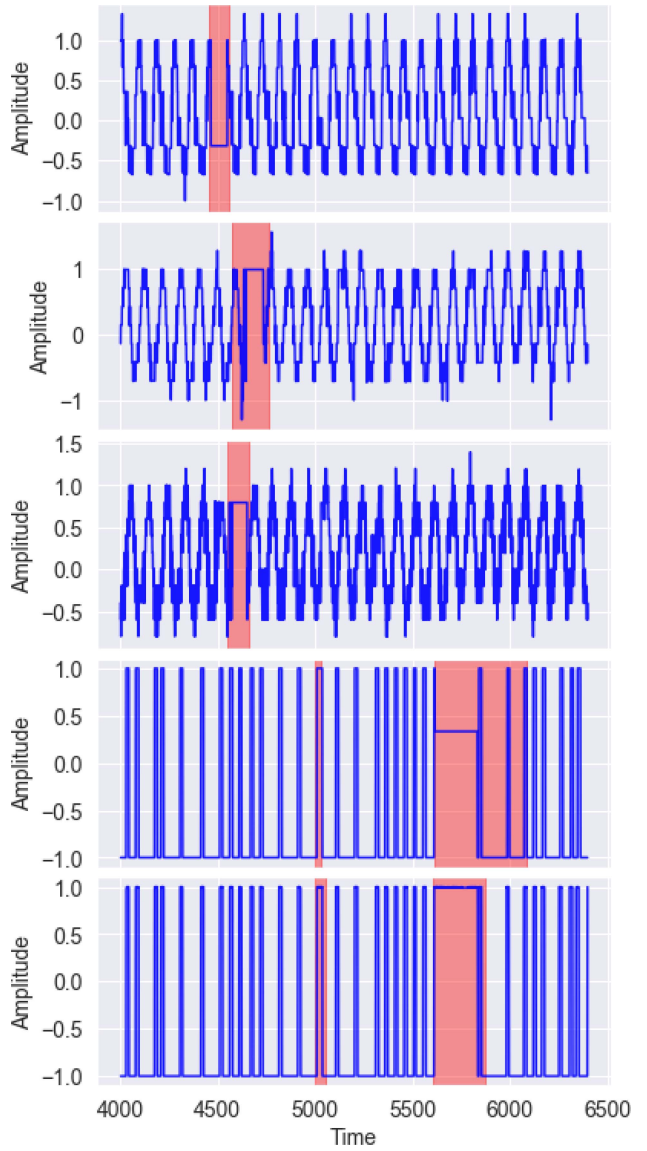


Fig. 10. Part of the test time series used in the case study.

anomaly sequences in all the groups. We treat each group of data as a ten-channel test case.

The SWaT dataset contains 51 channels of data collected in 11 days: seven days under normal operation and four days with attack scenarios. A total of 41 attacks appear in the last four days of data.

The WADI dataset contains 123 channels of data collected in 16 days: 14 days under normal operation and two days with attack scenarios, with 15 attacks launched during the last two days.

Following the work in [21] and [24], the original data samples for SWaT and WADI are downsampled to one measurement every 10 s by taking the median values. Since the attacks (anomalies) are labeled for the whole system rather than for every single channel in SWaT and WADI, after getting the absolute prediction error sequence for each channel, we add all of the error sequences together to generate an accumulated error sequence. We test the

Anomaly-MPP model on this 1-D sequence without the correlation prior.

B. Evaluation Metrics

Quantitative evaluation of the performance of anomaly detection in time series is not straightforward. There are mainly three ways commonly used to count anomalies: strict-point-based method, adjusted-point-based method, and segment-based method.

In the strict-point-based method, when a time point is classified as an anomaly, if it is labeled as a true anomaly in the ground truth, then this point is counted as a true positive (TP); otherwise, it is a false positive (FP). When a point is labeled as a true anomaly in the ground truth, but it is not classified as an anomaly in the detection, then it is a false negative (FN).

In the task of anomaly detection, the start and end points of an anomaly segment are sometimes “blurry” [48]. Often, partial detection of an anomaly is enough for experts to analyze the anomaly segment. Thus, the adjusted-point-based metric has been broadly applied to anomaly detection [28]. In the adjusted-point-based method, if any point of a true anomaly segment is detected as an anomaly point, then the whole true anomaly window is considered as correctly detected and every observation point in this segment will be counted as a TP. If no point of a true anomaly segment is detected as an anomaly, then all points belonging to this true anomaly segment are taken as FNs. The detected anomaly points outside the true anomaly segments are treated as FPs.

For the segment-based counting method [10], a detected anomaly segment consists of consecutive detected anomaly points. A TP is recorded if any portion of a true anomaly segment falls within any detected segment of anomalies. An FN is recorded if no detected segments overlap with a true anomaly segment. An FP is recorded if a detected segment does not overlap with any true anomaly segment.

After defining the TP, FP, and FN in different counting methods, the standard evaluation metrics, namely, precision, recall, and F1-score can be used to evaluate the performance of an anomaly detection approach, where

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (18)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (19)$$

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (20)$$

To compare the point-based metrics and segment-based metric, an example with ground truth (blue line) and detected anomalies (orange line) is shown in Fig. 9. The evaluation results for this example are given in Table I. Compared with the strict-point-based method, the adjusted-point-based method re-labels 60 FN points to TP as the second and the third detected anomaly segments are overlapped with two true anomaly segments. We can see that the adjusted-point-based metric is insensitive to short FNs. Even though two of the five true anomaly segments are

TABLE I
Comparison of Different Anomaly Counting Methods

Method	TP	FP	FN	Recall	Precision	F1
SPB	55	140	90	0.38	0.28	0.32
APB	125	140	20	0.86	0.47	0.61
SB	3	2	2	0.60	0.60	0.60
SSB	3	4	2	0.60	0.43	0.50

SPB: Strict-point based, APB: Adjusted-point based, SB: Segment based, and SSB: Split-segment based.

not detected, the recall is still as high as 0.86, which is not reasonable in our application, since it is important to discover the short anomaly segments for system diagnosis.

The segment-based method works on segments rather than points, so each anomaly segment is equally counted. The recall is 0.6 in the example, which is more reasonable for our application. However, by expanding the length of a detected anomaly segment, one could improve the segment-based recall without reducing the precision. The first detected anomaly sequence in Fig. 9 gives an example for this case. Although only a small portion of this detected anomaly is covered by the true anomaly segment, the other parts do not contribute to FPs, so the segment-based recall and F1 scores could be improved by simply extending each detected anomaly segment, without any penalty to the precision.

The segment-based method is preferred in our task, but to reduce the effect of the aforementioned problem, we propose a split-segment-based method for counting anomalies. The way to count TP, FP, and FN is the same as the segment-based method, but we will set a maximum length $L_{\max}$ for the detected anomaly sequence. When the length of the detected anomaly sequence is larger than $L_{\max}$, the detected anomaly is split into shorter anomaly sequences with length less or equal to $L_{\max}$ before counting anomalies. In the example of Fig. 9, we set $L_{\max} = 50$, so the first detected anomaly segment is split into three parts by the dashed red lines, which results in two more false alarms. Note that this adjustment is made only for computing performance metrics and does not affect the detected anomalies presented to the user.

In analyzing the results of our experiments, we mainly focus on the split-segment-based metrics, and the maximum length of sequence $L_{\max}$ is set to 1000. Meanwhile, we present the adjusted-point-based results just for reference, since that is often used in the literature. Since our method considers not only the amplitude of the error value but also the length of an anomaly object, a detected anomaly object could contain many points with low error values, which may result in low precision with the adjusted-point-based metrics. Thus, we extract the peaks from a detected anomaly segment as detected anomaly points for calculating the adjusted-point-based results.

TABLE II
Parameters of Anomaly-MPP Model

Group	Para.	Description
Data energy	G_d	Data gain, fixed at 0.25.
	T_d	Data reward threshold.
	K_d	Data reward coefficient.
Length prior energy	α	Length prior weight.
	G_l	Length gain, fixed at 4.
	T_l	Length reward threshold.
	K_l	Length reward coefficient.
	$b_{\max}$	Maximum length of object.
Correlation prior energy	β	Correlation prior weight.
	G_{co}	Correlation gain, fixed at 4.
	T_{co}	Correlation reward threshold.
	K_{co}	Correlation reward coefficient.
	d_{cor}	Expanded width.

C. Experiment Settings

A transformer-based predictor was used to generate the prediction error sequences. To keep consistency, the parameters of the predictor were set to be the same for all of our experiments. The input window size parameter $L = 150$, model dimension $d_{\text{model}} = 36$, value dimension $d_v = 36$, key dimension $d_k = 96$, and decoder input size $L_d = 6$. In the training process, Adam optimizer was used with initial learning_rate = 0.001. Early stopping was applied when validation loss did not decrease for 30 epochs, and the maximum training epoch was 300. The batch size was set to 128.

The 14 parameters of the proposed Anomaly-MPP model are listed in Table II. There are three groups of parameters, namely, data energy parameters, length prior energy parameters, and correlation prior energy parameters. The three gain parameters G_d , G_l , and G_{co} from each group, which are used to control the range of the independent variable of the reward function, were fixed to the values in Table II for all of our experiments. The minimum and maximum lengths ($b_{\min}$ and $b_{\max}$) of anomaly objects were set to 6 and 600, respectively. The expanded width d_{cor} for calculating the correlation prior energy was set to 300. The values of the other eight parameters (T_d , K_d , w_l , T_l , K_l , w_{co} , T_{co} , and K_{co}), whose values differ for the four datasets for which we present results, are discussed in the following section. Note that the fixed parameter values were determined experimentally to work well for all four datasets tested.

All the experiments were implemented by Python and PyTorch, running on a workstation with an Intel Core i7-5930 K CPU and two 24 Gb Titan RTX GPUs.

D. Case Study

An anomaly detection example is presented in this section to illustrate how each term in the energy function of the Anomaly-MPP model works. We select five channels (A-2, A-3, A-4, E-1, and E-10) of telemetry time series from the SMAP dataset as the test series for this illustration.

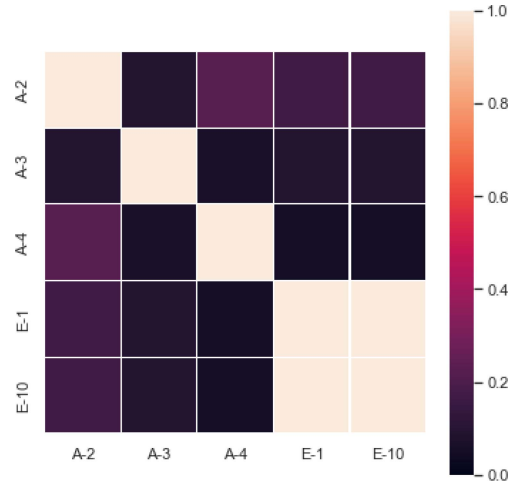


Fig. 11. Correlation heatmap of the test channels in the case study.

Part of the time series is shown in Fig. 10 and its absolute correlation heatmap is shown in Fig. 11. The two E-channels are highly correlated, the A-2 channel and the A-4 channel are weakly correlated, and A-3 seems not to be correlated with any other channels. There are a total of seven anomalies (red shaded segments in Fig. 10) in the test sequences of these five channels. The three anomalies in A-channels appear around time = 4500. Two short anomalies happen in E-channels around time = 5000, followed by two long anomalies around time = 5700.

Since the original time sequences are not aligned, we trained each channel separately with the transformer-based predictor. The absolute prediction error sequences are generated with the trained predictor. Fig. 12 shows all five of the absolute prediction-error sequences for this case study, superimposed. The Anomaly-MPP model aims to extract the anomalies from these error sequences.

1) *Data Energy With Overlap Prior Energy*: In the beginning, we only include the data energy and overlap prior energy. Thus, the weights for length prior and correlation prior are set to zero ($\alpha = 0$, $\beta = 0$). The overlap prior is kept to prohibit the overlapping of objects in the same channel. In this case, the fitting of objects is mainly determined by the data energy term $\hat{V}_d(e|w_i)$. When $\hat{V}_d(e|w_i) < 0$, the object w_i is more likely to be an anomaly. When w_i and w_j are overlapped in the same channel, the object with larger $\hat{V}_d$ will be killed. According to (6), and as seen in Figs. 5 and 6, the reward function h is monotonically decreasing with the independent variable x , the threshold parameter T controls when to reward, and the reward coefficient parameter K is related to the degree of reward only when $x > T$. For a fixed $K_d = 5$, we test with different T_d . Table III gives the test results with different T_d . When T_d is very small (e.g., $T_d = 0.8$), we can extract all the TP with a lot of FPs. When we increase T_d , the precision goes up, however, the recall may also decrease as in the case of $T_d = 2.4$. Since the data energy is based on the error contrast (between the inner and outer area of the object), generally, it is impossible to extract all the true anomalies without generating any false alarms.

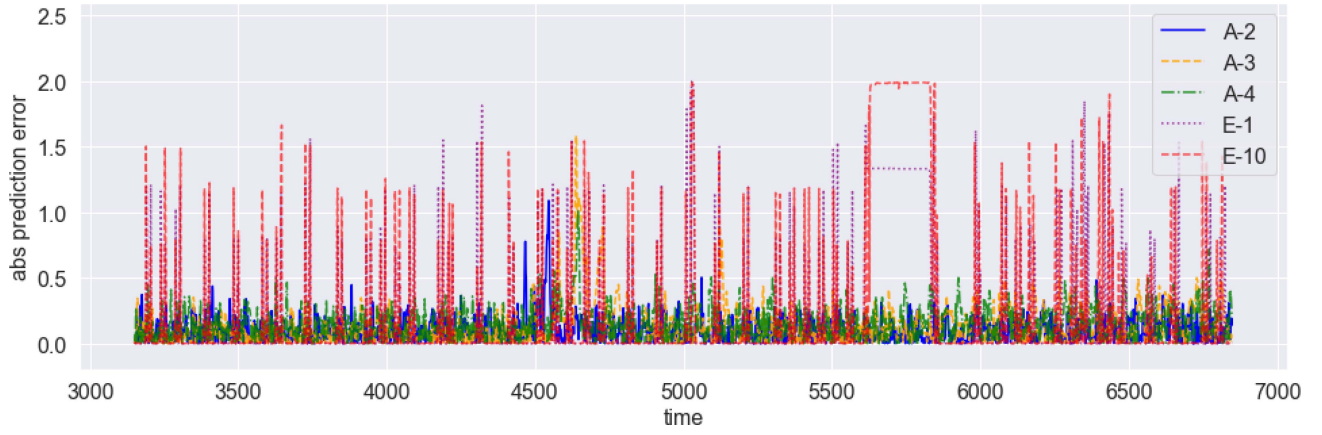


Fig. 12. Superimposed absolute prediction errors for each channel in the case study.

TABLE III
Extracted Results With Fixed $K_d = 5$ and Different T_d

T_d	S-Rec	S-Prec	S-F1	P-Rec	P-Prec	P-F1
0.8	1.0	0.045	0.086	1.0	0.689	0.816
1.2	1.0	0.07	0.131	1.0	0.779	0.876
1.6	1.0	0.108	0.195	1.0	0.851	0.92
2.0	1.0	0.152	0.264	1.0	0.9	0.947
2.4	0.857	0.231	0.364	0.959	0.949	0.954

Rec: Recall, Prec: Precision, S-: Split-segment based, and P-: Adjusted-point based.

TABLE IV
Extracted Results With Fixed $K_l = 5$ and Different T_l

T_l	S-Rec	S-Prec	S-F1	P-Rec	P-Prec	P-F1
20	1.0	0.28	0.437	1.0	0.915	0.956
40	1.0	0.875	0.933	1.0	0.99	0.995
60	1.0	0.875	0.933	1.0	0.988	0.994
80	0.857	1.0	0.923	0.91	0.989	0.948
100	0.571	1.0	0.727	0.844	0.99	0.911

Rec: Recall, Prec: Precision, S-: Split-segment based, and P-: Adjusted-point based.

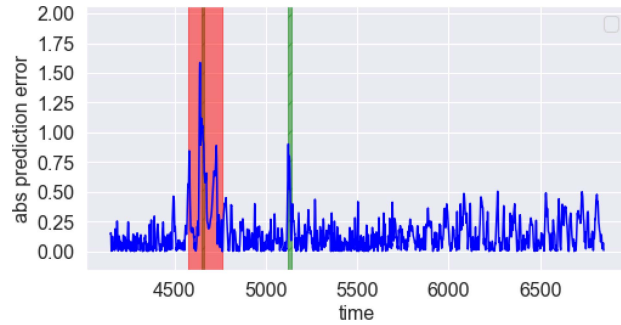


Fig. 13. Detecting results from channel A-4 with data energy and overlap energy. Red shaded area: ground truth, and green hatched area: detection results.

Fig. 13 shows the extraction results in channel A-4 when $T_d = 1.6$. In addition to the true anomalies, the detected false alarms also have high error contrast. They generally also have short lengths.

2) *Adding Length Prior Energy*: In this section, we fix $K_d = 5$ and $T_d = 1.6$ and add the length prior energy to our previous test model by setting the weight of length prior $\alpha = 0.15$ to penalize the short objects and encourage long objects. Again, we keep the length reward parameter $K_l = 5$ and try different T_l in our test. From the results given in Table IV, by introducing the length prior term, the precision improved in all of the five cases. The short FP in A-4 presented in Fig. 13 is killed in the case $T_l = 60$. However, it is still not easy to remove all false alarms with the added length prior energy.

TABLE V
Extracted Results With Fixed $K_{co} = 5$ and Different T_{co}

T_{co}	S-Rec	S-Prec	S-F1	P-Rec	P-Prec	P-F1
0.5	1.0	0.778	0.875	1.0	0.987	0.993
1.0	1.0	1.0	1.0	1.0	0.994	0.997
1.5	1.0	1.0	1.0	1.0	0.994	0.997
2.0	0.857	1.0	0.923	0.959	0.996	0.997
3.0	0.857	1.0	0.923	0.91	0.992	0.949

Rec: Recall, Prec: Precision, S-: Split-segment based, and P-: Adjusted-point based.

3) *Adding Correlation Prior Energy*: We may remove all the false alarms by fine-tuning the parameters of K_d , T_d , α , K_l , and T_l . However, the parameter tuning process for just a few false detections could be tedious and time-consuming due to the tradeoff between recall and precision. Instead of fine-tuning parameters, we can introduce some more prior knowledge to improve the detection results. Here, we add the correlation prior energy to the previous model by setting the weight of the correlation prior $\beta = 0.15$ and keeping $K_d = 5$, $T_d = 1.6$, $\alpha = 0.15$, $K_l = 5$, $T_l = 60$, and $K_{co} = 5$ in the new test. Also, we fix the neighbor distance $d_{cor} = 300$ for the correlation prior. The test results with different T_{co} are given in Table V. For the cases of $T_{co} = 1.0$ and $T_{co} = 1.5$, we reached the 1.0 F-1 score in the segment-based metric.

It is worth noting that the correlation prior is introduced to capture dependencies among anomaly objects occurring in different channels. This is particularly beneficial for scenarios where an anomaly appears in one channel, followed by anomalies occurring in other channels, which is a common case in spacecraft subsystems. When anomaly

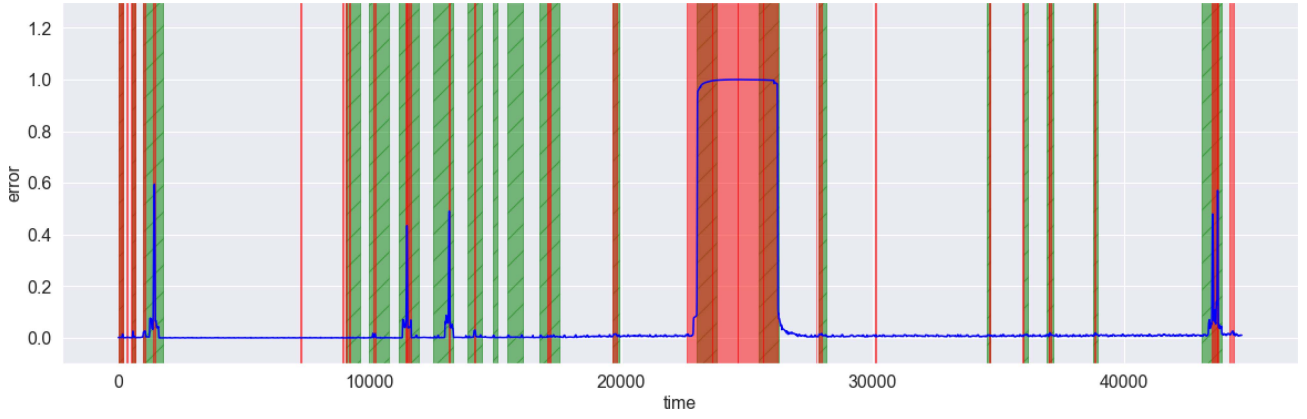


Fig. 14. Detection results of SWaT dataset. Red shaded area: ground truth, and green hatched area: detection results.

TABLE VI
Anomaly Detection Results of MRO-MUL Dataset

Method	S-Rec	S-Prec	S-F1	P-Rec	P-Prec	P-F1
DT	0.643	0.811	0.717	0.713	0.668	0.69
POT	0.85	0.763	0.804	0.884	0.693	0.777
MPP*	0.871	0.758	0.811	0.924	0.763	0.836
MPP	0.886	0.879	0.882	0.927	0.785	0.85

DT: Dynamic threshold, POT: Peaks-over-threshold, MPP: Anomaly-MPP, MPP*: Anomaly-MPP without correlation prior, Rec: Recall, Prec: Precision, S-: Split-segment based, and P-: Adjusted-point based.

objects in different channels are independent, it is preferable to conduct single-channel analysis by setting $\beta = 0$ to disregard the correlation prior energy.

E. Test on MRO-MUL Dataset

There are eight groups of data in MRO-MUL dataset, each containing ten channels. For each group of data, a transformer-based predictor is trained in a multichannel-in and multichannel-out manner. By applying the trained predictor on the test data, we can generate ten absolute error sequences for each group. Then, we extract the anomalies from the error sequences by several different anomaly extraction methods for comparison: POT [34], dynamic threshold (DT) [10], and Anomaly-MPP. For our proposed method, we set the same parameters for all of the eight groups of tests by trial and error: $T_d = 2.2$, $K_d = 5$, $\alpha = 0.2$, $T_l = 120$, $K_l = 5$, $\beta = 0.3$, $T_{co} = 3$, and $K_{co} = 6$. The quantitative test results are given in Table VI.

F. Test on SWaT and WADI Dataset

There are 51 channels of data in the SWaT dataset. We trained a transformer-based predictor on all the channels in a multi-in and multi-out manner. Absolute prediction error sequences are generated from the test data spanning four days. Unlike what we do in the test of MRO-MUL dataset, we do not extract anomalies from each channel separately. Instead, we treat the combination of the 51 channels as a single system since the true anomaly points are labeled at the system level rather than the channel level. We added the 51 absolute prediction error sequences into a single error

TABLE VII
Anomaly Detection Results of SWaT Dataset

Method	S-Rec	S-Prec	S-F1	P-Rec	P-Prec	P-F1
DT	0.395	0.882	0.542	0.780	0.897	0.834
POT	0.553	0.955	0.7	0.850	0.910	0.879
MPP*	0.816	0.939	0.873	0.852	0.905	0.878

DT: Dynamic threshold, POT: Peaks-over-threshold, MPP: Anomaly-MPP, MPP*: Anomaly-MPP without correlation prior, Rec: Recall, Prec: Precision, S-: Split-segment based, and P-: Adjusted-point based.

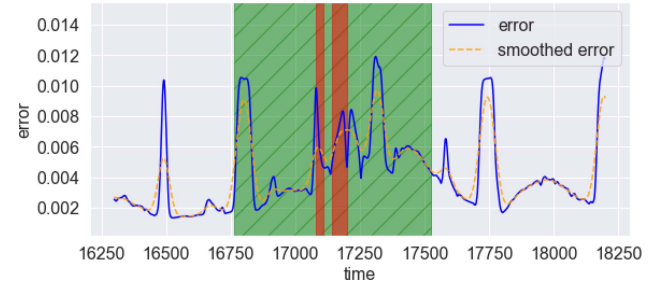


Fig. 15. SWaT detection results in the range of [16 300, 18 200]. Red shaded area: ground truth, and green hatched area: detection results.

sequence and scaled the error value into range [0, 1] as the blue line shown in Fig. 14. To extract the anomalies from this sequence, we reset the parameters of the Anomaly-MPP model as: $T_d = 0.4$, $K_d = 6$, $\alpha = 0.4$, $T_l = 120$, and $K_l = 1$. The correlation prior is not used, thus $\beta = 0$. The detected results are given in Table VII. Our method still outperforms the threshold-based anomaly extraction methods even though the target error sequence is single. To see why this happens, let us focus on the sample range [16 300, 18 200] of the error sequence, which is presented in Fig. 15. The maximum prediction error value in the true anomaly sequence is smaller than the errors in other normal places, which makes it impossible for a threshold-based method to extract the true anomalies without extracting other false alarms. But with the length prior in our model, the true anomaly is extracted without any false alarms because its accumulated error is larger than the accumulated error in other places and the length prior encourages longer anomaly objects while penalizing shorter anomaly objects. Note that

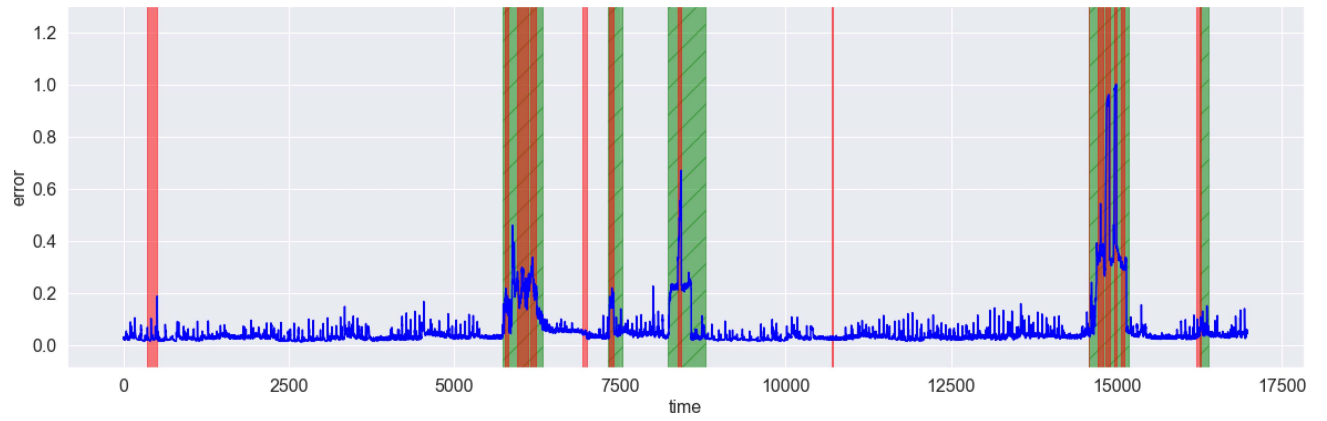


Fig. 16. Detection results of WADI dataset. Red shaded area: ground truth, and green hatched area: detection results.

TABLE VIII
Anomaly Detection Results of WADI Dataset

Method	S-Rec	S-Prec	S-F1	P-Rec	P-Prec	P-F1
DT	0.643	1.0	0.783	0.629	0.508	0.562
POT	0.643	1.0	0.783	0.689	0.603	0.643
MPP*	0.786	1.0	0.88	0.667	0.851	0.748

DT: Dynamic threshold, POT: Peaks-over-threshold, MPP: Anomaly-MPP, MPP*: Anomaly-MPP without correlation prior, Rec: Recall, Prec: Precision, S-: Split-segment based, and P-: Adjusted-point based.

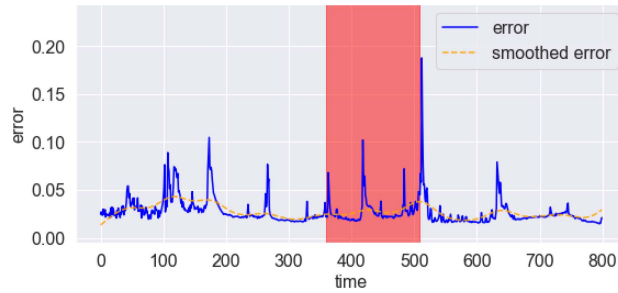


Fig. 17. One of the FNs in the WADI detection results. Red shaded area: ground truth, and green hatched area: detection results.

although shorter anomaly objects are penalized, it is still possible to detect short anomalies if the other energy terms in the model favor it enough.

Although there are a total of 123 channels in the WADI dataset, we processed it in the same way as we did for the SWaT dataset. The anomaly detection results are shown in Fig. 16 and Table VIII. Note that the recall for this dataset is not as high as for the other datasets. By checking the error sequence, there are no obvious clues for the missed anomalies. Fig. 17 shows the subsequence with one of the FNs. The error values in the true anomaly region are not significantly larger than the error values in other places, which resulted in the missed detection. This example shows that the extraction results are still highly dependent on the prediction results from the learning model since the data energy depends on the prediction errors.

G. Computational Considerations

Compared with the simple and fast thresholding-based anomaly extraction methods, the proposed Anomaly-MPP in the above experiments has slower processing speeds for different time series, with additional running times ranging from a few seconds to dozens of seconds. As an MPP optimized by MBD, the process of extracting anomalies by Anomaly-MPP involves simulating the birth (creation) and death (removal) of points (anomaly objects) in a point process, which require complex calculations. In addition, like many Monte Carlo simulation algorithms, MBD may require many iterations to converge to a stable solution, which can contribute to the algorithm's overall running time. To accelerate the proposed method, one potential approach is to carefully parallelize the birth and death operations across multiple processing units, although this is beyond the current scope of this work.

V. CONCLUSION

In this article, we proposed an anomaly detection method for multichannel time series, using a transformer-based predictor for generating prediction error sequences and an Anomaly-MPP model with MAP estimation for extracting the anomalies from the error sequences. The proposed Anomaly-MPP model is mainly controlled by three energy terms: the data energy, length prior energy, and correlation prior energy. The data energy encourages objects with high error contrast to be detected. The length prior energy suppresses short objects with low error contrast and rewards objects with longer lengths. The correlation prior energy introduces the information from other channels. A case study on the time sequences from the SMAP dataset is given to show the function of each energy term. Experimental results on MRO-MUL dataset, SWaT dataset, and WADI dataset prove the superiority of our Anomaly-MPP model over the threshold-based anomaly extraction methods that have been previously proposed.

Although our Anomaly-MPP model is quite flexible, the performance is still highly dependent on the error sequences generated by the predictor since the object data energy

is based on the error contrast. This is difficult to correct with a deep-learning approach for prediction, because of the “black-box” nature of deep learning systems. However, it is not necessary for the data energy to be defined on prediction error sequences. For example, we could define the data energy according to the variance of the values in the original input sequence. In our future work, we plan to explore some other data energy terms that do not depend on prediction error sequences.

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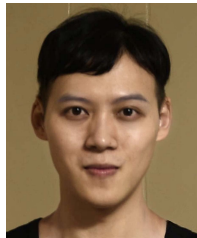
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